Large Language Models against Historical Ciphers: A Profiled Record of 296 European Targets, 1398–1945

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Abstract

We report what agentic large language models (LLMs) achieved on 296 European cipher targets dated 1398–1945, mostly diplomatic letters in archives, attacked between 14 September 2026 and 24 September 2026. Each target has a fixed profile: the cipher system, the documents, what the model started from, whether a reading already existed, the ordered solution steps, and an outcome split into *method* (how the text was obtained) and *extent* (complete or partial). The text was obtained for 211 of the 273 targets that could be worked (77 %), but that figure hides the methods. The key was recovered here for 67: 24 from the ciphertext alone, 34 from plaintext beside the cipher and 9 from plaintext found elsewhere. For 90 a key that already existed was found and applied. For 54 a decipherment already existed. From ciphertext alone the model recovered the key in 19 of 65 attempts. On the 76 targets with no known earlier reading, the text was obtained for 38. The model's main strength was locating keys, cribs and decipherments; its main failures were noisy transcriptions, short texts and large nomenclators without key material.

1 Introduction

Historical cryptology has been reshaped twice in the last two decades: by metaheuristic search, which made ciphertext-only attacks on homophonic and transposition ciphers routine (Lasry, 2018), and by shared collections such as DECODE, which put thousands of ciphertexts and keys in one place (Megyesi et al., 2019; Megyesi et al., 2020). Large language models add a third element. An agentic model can read a manuscript image, search archive catalogues and printed editions, write and run a solver, and judge whether its output is language. Whether that adds up to useful cryptanalysis, and on which ciphers, is an empirical question.

This paper answers it from a working record rather than a benchmark. Between 14 September 2026 and 24 September 2026 agentic models worked through lists of unsolved or unread historical ciphers: Tomokiyo's list of unsolved historical ciphers (Tomokiyo, 2026), Schmeh's list of the top fifty unsolved encrypted messages (Schmeh, 2017), the DECODE records marked non-decrypted, and a sweep of the manuscript catalogue of the Bibliothèque nationale de France. All of the work was done in one version-controlled repository, and that repository is the record we analyse: a folder per target holding the agent's working notes, the transcriptions, the solver code and the readings, and a commit history (1,982 commits over the eleven days) that dates every step and names the model that made it (Section 4). The weakness of such a record is that each target's notes describe its cipher in their own words. To make the targets comparable, we defined a fixed profile and filled it for every target from the notes and the files, measuring counts wherever they could be measured.

A record of this kind also forces a question the field has left informal: what counts as a success. A letter can be "deciphered" because the key was rebuilt from nothing, because a key sheet was found in another volume, or because a contemporary clerk had already written the plaintext between the lines. We therefore classify every outcome by method, following the categories proposed by the second author, and report the extent of the text obtained separately.

Our contributions are:

- a profile schema for historical cipher chal-

lenges and their solutions, in which the factors that plausibly affect success are recorded the same way each time (Section 3);

- profiles for 296 targets and 780 ciphertext documents from 1398 to 1945, in 13 plaintext languages (Section 5);
- an analysis of outcome method and extent by starting point, prior reading, cipher family and length (Section 6);
- six case studies, one per outcome route, and a discussion of the failures (Sections 7–8).

2 Related work

Ciphertext-only attacks on historical ciphers by hill climbing and simulated annealing are reviewed by Lasry (2018), whose methodology covers homophonic, transposition and machine ciphers; recent HistoCrypt and *Cryptologia* papers apply it to single nomenclators (Lasry, 2023) and to whole correspondences (Lasry et al., 2023). The scoring function in such attacks is an n-gram language model, and Megyesi et al. (2023) show that models built from period texts improve decipherment of seventeenth-century and older homophonic ciphers. The DECODE database (Megyesi et al., 2019; Megyesi et al., 2020) standardises the metadata of ciphertexts and keys (provenance, dating, symbol set, transcription); our profile borrows its document fields and adds fields about the attack and its result.

Neural models have been trained to decipher directly. Sequence-to-sequence models learn to crack simple substitution ciphers (Aldarrab and May, 2021), and a causal language model trained on symbol recurrences recovers keys of real homophonic ciphers (Kambhatla et al., 2023). Bayesian inference had earlier reached 97.8 % on the Zodiac-408 cipher (Ravi and Knight, 2011). These systems are cipher solvers; they do not find sources or decide what a document is.

General-purpose LLMs have been evaluated on ciphers as reasoning tasks. CipherBank (Li et al., 2025) tests chat and reasoning models on nine encryption algorithms applied to synthetic plaintexts and finds that even the strongest models solve fewer than half of tasks that human cryptanalysts can; Maskey et al. (2025) benchmark LLMs on decrypting ciphertexts of several classical and modern schemes. Both use generated ciphertexts with known answers and no archival context. We know of no evaluation of LLMs on archival ciphertexts, where most of the work lies before and around the cryptanalysis: finding the document, transcribing it, identifying the system, and finding the key or a decipherment if one survives.

3 The profile

A profile is a JSON file in five parts, validated against a schema. Every field may take the value *unknown*; a gap is then explicit and counted, rather than filled by a plausible guess.

System. The cipher types that occur (monoalphabetic, homophonic, polyphonic, syllabic, nomenclator, code, transposition, polygraphic, polyalphabetic, superencipherment; several at once means a combined system). The symbol kind (digits, letters, graphic symbols or mixtures), digit-group width and separation, and the number of distinct symbols. Diacritics and what they do. Homophones per vowel and per consonant. The nomenclator: presence, size, kinds of entry and the number of code groups still open. Nulls, word division and doubled letters. *Key order*: whether the key assigns its signs in alphabetical order, so that a few known values predict the rest. Any sibling key from the same office.

Documents. One entry per ciphertext unit: shelfmark, date, issuing power, route, plaintext language, pages. Whether the document contains clear text, and whether a plaintext exists and where: interlinear, in the margin, on a separate folio or in a separate collection, in a printed edition, or in an unrelated source. The length in enciphered tokens and the number of distinct tokens, flagged as measured when a common script counted them from the transcription file. Who transcribed the text (the model from images, a human, a published or DECODE transcription) and the image quality.

Conditions. Whether a reading of the text existed anywhere before this project obtained it, where, whether it was found before, during or after the attempt, and whether the work used it. This is the contamination question: a model may have seen a printed decipherment in training. The inputs the attack started from and the attack class (ciphertext-only, crib, known plaintext, archive decipherment, sibling key, key from archive, published key). The human's role, the tools, the model identifiers and the dates.

Solution. The ordered steps, each with a kind (access, transcription, statistics, hypothesis, crib, solver, control, sibling key, key from source, prediction, reading, verification, literature search) and a result (worked, partial, failed, ruled out). Failed steps are recorded.

Outcome method. How the text was obtained, in one of eight categories. Three recover a key here, by cryptanalysis: *key recovered from ciphertext only* (frequencies, solvers, pattern words, the order of the key; no plaintext and no existing key); *key recovered from adjacent plaintext* (a partial decipherment between the lines or in the margin, clear passages in the same letter, a deciphered copy in the same file); and *key recovered from external plaintext* (a printed edition, a calendar of state papers, a copy in another archive). Two apply a key that already existed: *external key matched* (a key for another correspondent or series, or a key sheet filed elsewhere, found to fit) and *known key applied* (a key already tied to the document or its correspondence). One transcribes: *existing decipherment* (a complete decipherment already existed; the work was transcribing the cipher and checking the decipherment against it). The last two are *not solved* and *not applicable* (not a cipher, or the item could not be obtained). Applying a known key is not mechanical when the key is polyphonic or variable-length, and the profile records such cases in a separate contribution field.

Extent. How much of the text was obtained: complete or partial. The field has no agreed threshold. DECODE marks each record decrypted, partly decrypted or non-decrypted, with the status "freely decided by the owner of the record" (Megyesi et al., 2019), and we met records marked partly decrypted that had been printed in full a century earlier. Computational work reports token accuracy and calls about 95–98 % solved without fixing a cut-off (Ravi and Knight, 2011). Historians call a letter deciphered when the running text reads and a few names or code groups stay open. We state our own rule. A target is *complete* when (1) at least 95 % of its cipher tokens give sense, counted from the reading file; (2) no remaining gap was left untried; (3) every document in it is covered, so that a group with one unread letter is partial; and (4) what stays open is scattered code groups or names, or pieces blocked from outside (no key material, too little text, illegible, or held only in the archive). Conditions (2) and (4) are ours. Open nomenclator codes are counted separately, since a letter can read at 99 % of tokens with many code entries unknown. Where the share of tokens with a value and the share that reads as sense differ, both are given.

The profiles were written by model agents instructed to source every value from the target's folder and to measure lengths with the common script; Section 9 reports how accurate they are.

4 Working method

The record. The agents kept a git repository with one folder per target. Of the 296 targets, 295 have a notes file that the agent maintained while working: what was fetched and from where, the transcription conventions, each hypothesis and solver run with its result, the controls, the literature searched, the reading, and what remains open with the reason. Solver code is in the folder for 205 targets; transcriptions and reading files sit beside it, with crops of the manuscript images where their terms allow it. A finished target was also written up for a public web page and entered in shared ledgers of results. The agents committed their work as they went, and 1,907 of the 1,982 commits name the Claude model that made them in a co-author line (the rest are merges, a few housekeeping commits by hand, and commits from the Codex harness, whose model is recorded in the profile instead), so the history fixes the order and date of the steps. The profiles of Section 3 were filled from these folders and this history.

Models and harness. The work was done by LLMs running as coding agents in terminal harnesses with file, shell and web access. Claude models (mainly Opus 5, Opus 5.5 and Fable 5.1) worked on 269 targets in Claude Code; OpenAI GPT-5 and GPT-6 models worked on 16 targets in the Codex harness, 2 of them shared with a Claude model; the model is not recorded for 13. Long tasks were split among parallel sub-agents, for example one per manuscript page for transcription, and merged by the main agent. The corpus is too unbalanced between model families to compare them, and we do not.

What the humans did. The first author's part was small and mostly at the project level. The first author obtained a DECODE login with the privileges needed to see images and keys and

handed the agent the session cookie, so that it could fetch records itself; set up the harness; and posed each problem as a standing goal, which keeps the agent working until the stated condition holds. During the work the first author occasionally chose which of several leads the agent had identified to pursue. The first author did not transcribe, propose keys or readings, or write code. Arya Sanketbhai Patel worked 8 targets with Claude Code and contributed them to the repository as pull requests, which the first author reviewed and merged; the results on those targets are his, and they are marked † in the tables and the appendix: herbault1626 (existing decipherment, complete); hesse1603 (known key applied, complete); lorraine1592 (key from ciphertext only, partial); maisse1592 (existing decipherment, complete); nevers1589 (not applicable); poyntz1527 (existing decipherment, complete); sessa1593 (known key applied, partial); sormano1529 (known key applied, partial). His notes record the agent's steps, not his own part in them, so for these targets the human's role is not itemised. In 5 targets the first author went further and supplied a specific source (a printed book the agent could not download) or suggested a route (a candidate key, archive addresses); these are marked in the data. The agents did the rest: locating images through archive interfaces, transcription, statistics, hypotheses, writing and running solvers, searching printed editions, and writing up.

Tools. Solvers were written per target, mostly in Python with C# for speed: hill climbing and simulated annealing over homophonic and syllabic keys, annealers constrained to an alphabetical key design, monotone alignment of ordered key slots, hard EM aligners from ciphertext to a contemporary decipherment, and SAT encodings. The scoring models were character n-gram models built from period corpora, for example nineteenth-century editions of sixteenth-century French correspondence, shared across targets through one registry.

Conventions. Three project rules bear on the numbers. A negative result is reported only alongside a matched control: a synthetic ciphertext of the same length, alphabet and design that the same solver recovers, or fails to recover in the same way. Every reading is graded by its source: a primary key, a known-plaintext letter, uncertain, or inferred from context or from the alphabetical position of a sign. And before a target was treated as unsolved, the nineteenth-century editions of the correspondence and the DECODE records of the same volume were searched.

outcome method	n	complete	partial
key from ciphertext only	24	14	10
key from external plaintext	9	6	3
key from adjacent plaintext	34	16	18
key recovered, all routes	*67*	*36*	*31*
external key matched	12	7	5
known key applied	78	50	28
existing decipherment	54	51	3
not solved	62	–	–
not applicable	23	–	–
all targets	296	144	67

Table 1: Outcome method and extent for the 296 targets. The first three methods recover the key here; the next two apply a key that already existed; the sixth transcribes and checks a decipherment that already existed. Complete means the read bar of Section 3 is met.

century	n	plaintext language	n
14th c.	1	French	93
15th c.	13	English	53
16th c.	146	Italian	52
17th c.	59	Spanish	33
18th c.	45	German	26
19th c.	13	Latin	17
20th c.	7	Dutch	11
undated	12	Hungarian	4

Table 2: The corpus by century of the earliest document and by plaintext language (a target with letters in two languages counts under both; 13 languages in all).

5 The corpus

The corpus is restricted to classical ciphers of European origin dated between 1350 and 1950, the range covered by the DECODE collection; the project's work on material outside that range or origin is not reported here. Table 2 summarises the 296 targets. They hold 780 ciphertext documents, from 1398 to 1945. Most are diplomatic letters of the sixteenth (146) and seventeenth (59) centuries: French, Spanish, Italian, English, Imperial and papal embassy traffic, with 13 targets from the fifteenth century and 20 from the nineteenth and twentieth, including telegrams and military ra-

starting point	*n*	recovered	applied	existed	not solved
ciphertext only	65	19	0	4	42
crib from clear text	26	19	2	1	4
decipherment or known plaintext	42	16	5	20	1
a key (archive, print, sibling)	104	12	81	5	6
none recorded	36	1	2	24	9
all	273	67	90	54	62

Table 3: Outcome by what the attack started from (profile field conditions.attack). Columns: key recovered here (from ciphertext only, adjacent or external plaintext); key applied (a known key, or an external key matched to the document); a decipherment already existed; not solved. Targets classed not applicable are left out.

cipher family	*n*	recovered	applied	existed	not solved
homophonic or simple + nomenclator	133	35	54	28	16
substitution, no nomenclator	38	19	10	6	3
undetermined	33	1	2	6	24
code or codebook	32	5	9	7	11
syllabic nomenclator	30	7	13	5	5
transposition, polygraphic, polyalphabetic	7	0	2	2	3
all	273	67	90	54	62

Table 4: Outcome by cipher family, from the system types in the profile. Columns as in Table 3.

prior reading	*n*	recovered	applied	existed	not solved
none known	76	24	14	0	38
existed, known before the attempt	93	16	49	25	3
existed, found during the attempt	70	15	22	27	6
existed, found after or not at all	12	9	3	0	0
existed, when found unknown	3	0	1	2	0
unknown	19	3	1	0	15
all	273	67	90	54	62

Table 5: Outcome by whether a reading of the text existed anywhere before this project, and when it was found (the contamination question). Columns as in Table 3.

ciphertext tokens	*n*	recovered	applied	existed	not solved
< 250	69	13	15	12	29
250–999	68	18	21	11	18
1,000–4,999	55	15	26	6	8
$\geq 5{,}000$	23	8	14	0	1
unknown	58	13	14	25	6
all	273	67	90	54	62

Table 6: Outcome by ciphertext length (enciphered tokens summed over the documents of a target). Columns as in Table 3.

diograms. Another 12 targets have no established date but come from early-modern archival contexts. The main plaintext languages are French 93, English 53, Italian 52, Spanish 33, German 26, Latin 17.

A length is known for 218 targets (measured by script on every document for 149); together they hold 802,054 enciphered tokens. The agent transcribed 513 of the documents (66 %) itself from archive images; 149 came from DECODE transcriptions and 59 from published ones.

In all, 23 targets are classed not applicable: 11 turned out not to be ciphers (clear text with a cipher instruction, shorthand, key sheets, dummy numbers) and 9 could not be obtained because the manuscript was not digitised or not reachable. The analysis below uses the 273 targets that were worked.

6 Results

Overall. Table 1 gives the outcome for every target. The text was obtained for 211 of the 273 worked targets: 144 complete and 67 partial; the remaining 62 were not solved. The methods behind the 211 differ widely. The key was recovered here for 67 targets (36 complete): 24 from the ciphertext alone, 34 from adjacent plaintext and 9 from external plaintext. For 90 targets an existing key was applied: 78 with a key already tied to the correspondence and 12 with a key found elsewhere and matched to the document. For 54 targets a complete decipherment already existed, and the contribution was a transcription, a check of the decipherment, or a correction of the catalogue. Counting only the first three methods as cryptanalysis, the model recovered 67 keys in 273 attempts.

Starting point dominates. Table 3 sorts the worked targets by what the attack started from. With a key in hand, from an archive, a printed edition or a sibling letter of the same office, the text was obtained in 98 of 104 cases (94 %). With a contemporary decipherment of some of the letters, it was obtained in 41 of 42; most of these rebuilt the key from the decipherment and applied it to the undeciphered letters. With a crib from the clear text around the cipher, it was obtained in 22 of 26. From the ciphertext alone, the model recovered the key in 19 of 65 attempts (29 %); in 4 more, a decipherment turned up during the attempt, and 42 were not solved. The other 5 ciphertext-only recoveries began from a crib guessed from the subject of the letter or from the plaintext vocabulary of a sibling key, which uses no plaintext of the document itself.

The ciphertext-only recoveries. Table 7 lists the 24 targets whose key was recovered from the ciphertext alone. Most are homophonic or simple substitutions, some with a small nomenclator, on a few hundred to a few thousand tokens. The

target	year	language	system	tokens	key order	extent
siena1421	1420	Italian, Latin	homophonic or simple + nomenclator	13237	random	partial
r1854	1428	Italian, Latin	homophonic or simple + nomenclator	4694	not applicable	partial
soria1523	1523	Spanish	homophonic or simple + nomenclator	–	unknown	partial
rangone1530	1529	Italian	substitution, no nomenclator	264	unknown	partial
joachim1530	1530	Italian	homophonic or simple + nomenclator	402	unknown	complete
guiche1551	1551	French	substitution, no nomenclator	406	unknown	partial
memoire1560	1560	French	code or codebook	1715	unknown	complete
harley1582r8505	1585	French	substitution, no nomenclator	3564	random	complete
segur	1585	French	syllabic nomenclator	594	alphabetical	complete
lorraine1592†	1592	French	homophonic or simple + nomenclator	1109	unknown	partial
stowe166	1592	English	homophonic or simple + nomenclator	–	unknown	partial
segura1596	1596	Spanish	substitution, no nomenclator	125	unknown	complete
ferdinand1619	1619	German	homophonic or simple + nomenclator	2614	partly alphabetical	complete
warsaw	1627	Italian	syllabic nomenclator	465	alphabetical	complete
harley286	1635	Latin	substitution, no nomenclator	281	random	complete
marburg1635	1635	German	substitution, no nomenclator	590	unknown	complete
wesselenyi1664	1664	Latin	homophonic or simple + nomenclator	244	alphabetical	complete
potocka1714	1714	French, Latin, Polish	homophonic or simple + nomenclator	1119	partly alphabetical	partial
kauderbach1754	1754	French	homophonic or simple + nomenclator	8512	random	complete
swieten1757	1758	French	homophonic or simple + nomenclator	642	alphabetical	complete
r1892	1795	Dutch, French	substitution, no nomenclator	1872	random	partial
soglia1848	1848	Italian	code or codebook	955	alphabetical	complete
egmond	–	French	substitution, no nomenclator	774	unknown	complete
sp106r927	–	French	substitution, no nomenclator	–	unknown	partial

Table 7: The 24 targets whose key was recovered from the ciphertext alone. † worked by Arya Sanketbhai Patel.

key order is known for 7 of them to be alphabetical or partly alphabetical and for 5 to be random; an alphabetical key let the model predict unseen values and confirm them in the text, and in one case (Warsaw 1627) the key's alphabetical order, which the solver did not reward, was the independent check. All but 2 of the 24 carry a verification beyond the reading itself: a matched control, key consistency, or historical consistency with the known events. Of the 24, 12 had a reading somewhere that was found after the key was rebuilt or that covered only part of the target (3 known before the attempt, covering other letters or passages than those whose key was rebuilt); they are independent re-solutions and are marked as such in the data.

Prior readings. Table 5 splits the worked targets by whether a reading existed anywhere before the project, and when it was found. Where none is known (76 targets), the text was obtained for 38, 25 of them complete: 24 by recovering the key (12 from the ciphertext alone) and 14 by applying a key that had never been used on these letters. This is the subset free of contamination, since the model cannot have memorised a decipherment that does not exist. Where a reading existed, it was most often found by the agent during the attempt (70 targets) rather than known beforehand (93); 44 targets had a reading in print, usually in a nineteenth-century edition that the modern lists of unsolved ciphers had missed.

Cipher family matters less. By family (Table 4), homophonic ciphers with nomenclators are the largest group and were obtained most often, but most of those attacks started from a key. Simple and homophonic substitutions without a nomenclator have the most key recoveries relative to their number. The *undetermined* row, targets whose system was never established, is mostly not solved by construction. The comparison across families is confounded by starting point, and the cells are small.

Length. Short ciphertexts (under 250 tokens) were most often not solved (Table 6); above that, the share not solved falls with length, and almost no text of 5,000 tokens or more went unsolved. Long texts are mostly nomenclator letters for which a key was found or rebuilt from sibling decipherments.

Effort. The profiles hold 2,686 solution steps, of which 658 (24 %) failed or were ruled out. There are 251 solver runs, of which 170 failed or were ruled out, 376 literature searches and 306 transcription steps. Matched controls appear in 96 targets. The balance of steps reflects the work: searching the literature and the archives was as frequent as running solvers, and more often decisive.

What else the work added. Beyond the text, the profiles record corrections to the catalogues (124 targets: a wrong sender, date, or decryption status in DECODE or on a list of unsolved ciphers), identifications of an unknown sender or date (112), keys extended or corrected (65), and 14 decipherments with a polyphonic or variable-length key where applying a known key required judgement.

7 Case studies

Toledo to Philip II, 1565 (key from adjacent plaintext). A letter of the viceroy of Sicily during the siege of Malta, with 1,937 two-digit figures inline in clear Spanish (Archivo General de Simancas, Estado leg. 1394). The agent worked out the image route of the Spanish archives portal, transcribed the figures, and took two runs next to clear words as cribs (*seiscientos*, *en tiera*). They fixed eleven letters, and the values came out in alphabetical order, so the remaining letters were predicted from their neighbours and confirmed in words. Nonsense in the decode located misread digits, which were corrected on the images. A companion letter of the same day, found by searching the portal, decodes with the same key. No earlier decipherment is known.

Balbases to Fuenmayor, 1677–78 (key from adjacent plaintext). Fourteen Spanish despatches from the peace congress at Nijmegen, ten with a contemporary decipherment in the margin. Eight keys of the same series were tried and none fitted. Two sub-agents transcribed every cipher and margin line; anchors from n-grams repeated across letters seeded a hard-EM aligner that maps each token to a substring of the margin text, and the table converged in two rounds. It reproduces the ten margins and reads the four letters that carry no decipherment, 97 % of tokens.

Feuquières to Catinat, 1691 (key from external plaintext). A two-part code of 418 groups (160 distinct) in the "petit chiffre" of the governors of Pignerol, deciphered by Étienne Bazeries in 1893 without publishing the table. A matched control of the inferred design showed that annealing could not recover it: wrong keys outscored the true one. The agent then searched the digitised text of the 1819 *Mémoires* of Catinat for runs of numbers and found a letter of Louvois of 6 September 1690 printed in the same code with its translation. Anchoring by hand and propagating between the two letters read 586 of 601 tokens; the Feuquières text agrees with Catinat's own memoir of the same days.

Bordeaux to Brienne, 1653 (external key matched). A French despatch intercepted in London: 813 tokens of graphic signs and numbers, three number series marked by a prime, two dots or an overbar. From sibling keys of the same office the agent inferred the design (alphabetical syllabaries in the marked series) and built an annealer with that constraint. It recovered 99 % of matched controls and failed on the real text; controls with 10 % of the diacritics misread failed in the same way (52 % and 9 %). The agent then found the English Deciphering Branch key sheet for this correspondent in a separate volume of the British Library (DECODE R7537). It confirmed the design exactly, and a check against the images made 21 corrections to the transcription, splitting two merged glyphs and restoring lost marks. The key applied to the corrected transcription read the letter except name codes and rare tokens.

Henry of Navarre to Ségur, 1585–86 (key from ciphertext only). Figures in three letters, 466 tokens. Before any value was known, the frequencies of the upper figures taken modulo five fell into blocks of five, the signature of an alphabetical syllabary of consonants by vowels. An annealer constrained to that design, with the clear French around the cipher as fixed context, converged on a key and recovered 96.6 % of matched controls under random keys of the same shape. A sibling key published by Tomokiyo, with different offsets, read a fourth letter; the constrained annealer then re-derived that fourth key blind at 93 % of letter tokens, against 90 % on a matched control.

Ordnungspolizei radiograms, 1942 (not solved). Seven parts of 27 February 1942 in the German police Doppelkasten, a double Playfair in which each pair passes twice through the same two boxes, 970 letters under one key. The agent implemented the system from the German manuals and attacked it with annealing, iterated local search, tempering, relaxation and SAT. All of these failed on synthetic texts of the same length with a known key: the double pass leaves a local search no gradient. A known-plaintext solver it wrote recovers the boxes from 80 to 111 plaintext pairs on synthetic data, but no plaintext of these messages is available.

8 Where the models failed

Transcription. The largest single cause of failure was the transcription, not the cryptanalysis. The agent transcribed every document of 181 worked targets from images and obtained the text for 136 of them, but the Bordeaux case shows how a transcription that looks usable can defeat a correct solver. Controls with injected transcription noise are cheap and diagnostic, and we recommend them as a standard step.

Short texts and large nomenclators. Large nomenclators on a few hundred tokens were beyond reach without a key or a crib, as they are for human analysts. The agent's value here was in finding the key: on the DECODE record of a sibling letter, in the same archive volume, or in a printed edition.

Undetermined systems. Where the system was never identified, the target was almost never solved. These are mostly short texts, from a few lines of figures to a few hundred tokens: an isolated slip, a postscript, a fragment of a series whose other parts are lost.

Double encipherment and hand systems of the twentieth century. The double Playfair and the other twentieth-century hand systems in the corpus were not solved. The agents wrote correct implementations and competent searches and reported the negative results with controls.

Finding what was already known. Many targets on current lists of unsolved ciphers had been read long ago, most often in a nineteenth-century printed edition, sometimes in a decipherment bound in the next folio. The agent found these by searching the editions and the neighbouring archive records, and we classify them by the method that applied, never as key recoveries.

9 Threats to validity

The profiles are written by the agents. They were compiled by model agents from the project's notes, partly during the work and partly afterwards, and the key order is unknown for 187 targets, usually because the key was never recovered. The human's role is recorded at the project level for most targets, since the notes rarely mention it; lead choices within a session are not itemised. To estimate accuracy, ten profiles were drawn at random (fixed seed) from the first 82 and checked field by field against the notes, the result tables and the ciphertext files by fresh agents that had not written them. Of 249 fields checked, 38 were unknown. Of the 211 with a value, 196 agreed with the sources (93 %), 5 contradicted them and 10 asserted something the sources do not establish. All recorded lengths re-measured exactly. The errors were of three kinds: placeholder values in profiles of targets never obtained; a modern partial reading filed as a plaintext location; and over-confident system fields for ciphers that were never broken. They were corrected, and the schema gained a *margin* value for plaintext location. The audit was done by agents, not by a human, which limits what it shows.

The outcome method. The method was added to every profile after the work, by review agents that read each target's notes. A rule-based proposal computed from the older profile fields (attack class, prior solution, solution steps) agreed with the reviewers' method for 225 of 290 targets (78 %). The proposal flagged 115 targets as ambiguous, typically whether a sibling key had already been tied to the document (key applied) or had to be matched to it (external key), and whether a key came from plaintext beside the cipher or from plaintext elsewhere; 37 of the 65 disagreements fell among those. The boundary between *known key applied* and *external key matched* is the least certain; the boundary between recovering a key and applying one is well supported by the notes.

The extent rule is ours. The 95 % threshold follows computational practice, not a published standard. A numeric share of tokens read is recorded for 148 of the 211 targets with text (79 measured from the reading files, the rest estimated from the notes). Raising the threshold to 98 % would move at most 29 complete targets to partial; lowering it to 90 % would move at most 12 partial targets to complete. Another 4 targets read above 95 % but are partial for another reason, usually a document of the group left unread.

Selection. The targets came from lists of unsolved ciphers, from DECODE records marked non-decrypted and from a catalogue sweep, and the agent chose the order of attack. Targets that looked hopeless were attempted briefly or set aside, so the not-solved count mixes serious failures and brief closures.

No fixed budget. Sessions varied in length, model and operator, and the first author supplied logins and in a few cases a source. The record is of what happened, not of a controlled experiment.

Contamination. A model may have seen a printed decipherment in training. The prior-reading field addresses this directly, and the 76 targets with no known prior reading are the clean subset. For the others, the profile records whether the work used the prior reading, and the method records whether the key was rebuilt or applied.

10 Conclusion

On historical ciphers, an agentic LLM behaves less like a solver than like a research assistant that can also write solvers. It reads the manuscript, finds the key or the decipherment in the archive or in print, recognises a crib in the clear text, and exploits structure such as an alphabetical key. Of the 211 targets whose text it obtained, the key was recovered here for 67, and from the ciphertext alone for 24. Its failures follow the classical limits: short texts, large nomenclators, double encipherment and noisy transcriptions. Classifying outcomes by method keeps these two contributions apart, and we recommend it to anyone reporting decipherments made with such tools. The profile schema, the 296 profiles and the working record behind them (notes, transcriptions, code and commit history) are released with the paper so that later work can check any profile against its sources and add targets and models on the same terms.

Acknowledgements

Arya Sanketbhai Patel worked the 8 targets marked †, including the ciphertext-only key recovery for the letter of Charles III of Lorraine to Vaudémont (1592), and wrote them up for the project site.

References

Nada Aldarrab and Jonathan May. 2021. Can sequence-to-sequence models crack substitution ciphers? In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pages 7226–7235.

Nishant Kambhatla, Logan Born, and Anoop Sarkar. 2023. Decipherment as regression: Solving historical substitution ciphers by learning symbol recurrence relations. In *Findings of the Association for Computational Linguistics: EACL 2023*, pages 2136–2152.

George Lasry, Norbert Biermann, and Satoshi Tomokiyo. 2023. Deciphering Mary Stuart's lost letters from 1578–1584. *Cryptologia*, 47(2):101–202.

George Lasry. 2018. *A Methodology for the Cryptanalysis of Classical Ciphers with Search Metaheuristics*. Kassel University Press, Kassel.

George Lasry. 2023. Armand de Bourbon's polyhomophonic cipher – 1649. In *Proceedings of the 6th International Conference on Historical Cryptology (HistoCrypt 2023)*, pages 105–112. Linköping University Electronic Press.

Yu Li, Qizhi Pei, Mengyuan Sun, Honglin Lin, Chenlin Ming, Xin Gao, Jiang Wu, Conghui He, and Lijun Wu. 2025. CipherBank: Exploring the boundary of LLM reasoning capabilities through cryptography challenge. In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 5929–5965, Vienna, Austria.

Utsav Maskey, Chencheng Zhu, and Usman Naseem. 2025. Benchmarking large language models for cryptanalysis and side-channel vulnerabilities. arXiv:2505.24621.

Beáta Megyesi, Nils Blomqvist, and Eva Pettersson. 2019. The DECODE database: Collection of historical ciphers and keys. In *Proceedings of the 2nd International Conference on Historical Cryptology (HistoCrypt 2019)*, pages 69–78.

Beáta Megyesi, Bernhard Esslinger, Alicia Fornés, Nils Kopal, Benedek Láng, George Lasry, Karl de Leeuw, Eva Pettersson, Arno Wacker, and Michelle Waldispühl. 2020. Decryption of historical manuscripts: the DECRYPT project. *Cryptologia*, 44(6):545–559.

Beáta Megyesi, Justyna Sikora, Filip Fornmark, Michelle Waldispühl, Nils Kopal, and Vasily Mikhalev. 2023. Historical language models in cryptanalysis: Case studies on English and German. In *Proceedings of the 6th International Conference on Historical Cryptology (HistoCrypt 2023)*. Linköping University Electronic Press.

Sujith Ravi and Kevin Knight. 2011. Bayesian inference for Zodiac and other homophonic ciphers. In *Proceedings of the 49th Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 239–247.

Klaus Schmeh. 2017. The top 50 unsolved encrypted messages. Klausis Krypto Kolumne, `https://scienceblogs.de/klausis-krypto-kolumne/the-top-50-unsolved-encrypted-messages/`.

Satoshi Tomokiyo. 2026. Unsolved historical ciphers. `https://cryptiana.web.fc2.com/code/unsolved.htm`.

Table 8: Every target in the corpus, by date of the earliest document. Methods: CT key from ciphertext only; EPT key from external plaintext; APT key from adjacent plaintext; EK external key matched; KK known key applied; DEC existing decipherment; NS not solved; NA not applicable. † worked by Arya Sanketbhai Patel. Full profiles are in the data release.

target	year	language	system	method	extent	tokens
benedict1399	1398	Catalan, Latin	homophonic or simple + nomenclator	DEC	complete	–
florence1414	1414	–	substitution, no nomenclator	NA	n/a	–
siena1421	1420	Italian, Latin	homophonic or simple + nomenclator	CT	partial	13237
florence1429	1425	Italian	homophonic or simple + nomenclator	KK	complete	99
medici1425	1425	Italian	undetermined	NA	n/a	–
heidelberg597	1426	German	substitution, no nomenclator	DEC	complete	–
r1854	1428	Italian, Latin	homophonic or simple + nomenclator	CT	partial	4694
maria1435	1435	Catalan	undetermined	NS	none	42
it1583	1446	Italian	homophonic or simple + nomenclator	NS	none	794
beatrice1482	1482	Italian	undetermined	DEC	complete	–
matthias1482	1482	Latin	homophonic or simple + nomenclator	DEC	complete	330
sadoleto1482	1482	Italian	homophonic or simple + nomenclator	APT	partial	–
buda1489	1489	Italian	syllabic nomenclator	KK	partial	469
esp318	1497	Italian, Spanish	homophonic or simple + nomenclator	KK	partial	1979
estrada1503	1503	Spanish	homophonic or simple + nomenclator	KK	partial	–
vich1511	1511	Spanish	syllabic nomenclator	APT	partial	25736
bonzagni1512	1512	Italian	substitution, no nomenclator	DEC	complete	–
ayala1516	1516	Spanish	homophonic or simple + nomenclator	KK	complete	–
schiner1516	1516	Latin	undetermined	DEC	complete	–
jantini1517	1517	Italian	undetermined	NS	none	581
r4349	1517	Latin, Swedish	substitution, no nomenclator	KK	partial	131
spinelly1516	1517	English	homophonic or simple + nomenclator	APT	complete	43
caprile1519	1519	Italian	substitution, no nomenclator	EPT	partial	2284
adrian1521	1521	Spanish	syllabic nomenclator	EPT	complete	–
lopehurtado	1522	Spanish	homophonic or simple + nomenclator	APT	partial	29
nino1522	1522	Spanish	substitution, no nomenclator	NS	none	128
poupet1522	1522	French	substitution, no nomenclator	DEC	complete	–
r9658	1522	Spanish	homophonic or simple + nomenclator	DEC	complete	–
sanchez1522	1522	Spanish	homophonic or simple + nomenclator	KK	partial	33
harley6345	1523	English	homophonic or simple + nomenclator	NA	n/a	78
lopehurtado1523	1523	Spanish	homophonic or simple + nomenclator	APT	partial	–
sanchez1523	1523	Spanish	homophonic or simple + nomenclator	KK	complete	–
sessa1523	1523	Spanish	homophonic or simple + nomenclator	EK	partial	–
soria1523	1523	Spanish	homophonic or simple + nomenclator	CT	partial	–
moncada1524	1524	Spanish	homophonic or simple + nomenclator	EPT	complete	442
sessa1524	1524	Spanish	homophonic or simple + nomenclator	APT	partial	–
mellon29	1525	–	undetermined	NS	none	234
bayard1526	1526	French	undetermined	NS	none	56
lee1526	1526	English	homophonic or simple + nomenclator	KK	complete	823
raince	1526	French	homophonic or simple + nomenclator	KK	partial	5629
poyntz1527†	1527	English	homophonic or simple + nomenclator	DEC	complete	44
soria1527	1527	Spanish	homophonic or simple + nomenclator	DEC	complete	–
vasto1527	1527	Italian, Spanish	code or codebook	KK	partial	3001
visconti1528	1528	Italian	homophonic or simple + nomenclator	KK	partial	1344
visconti1529	1528	Italian	homophonic or simple + nomenclator	KK	complete	1672
bizozola1520	1529	Italian	substitution, no nomenclator	KK	complete	1914
dubellay	1529	French	homophonic or simple + nomenclator	DEC	complete	–
gramont1529	1529	French	homophonic or simple + nomenclator	KK	complete	4002
kaa4591	1529	German, Latin	undetermined	APT	partial	–
rangone1530	1529	Italian	substitution, no nomenclator	CT	partial	264
sormano	1529	Italian	homophonic or simple + nomenclator	KK	complete	–
sormano1529†	1529	Italian	substitution, no nomenclator	KK	partial	–
joachim1530	1530	Italian	homophonic or simple + nomenclator	CT	complete	402
isabella1531	1531	Spanish	syllabic nomenclator	DEC	complete	–
kaa4591b	1531	German, Latin	homophonic or simple + nomenclator	APT	partial	9468
mai1531	1531	Spanish	undetermined	NA	n/a	–
muxetula1531	1531	Spanish	homophonic or simple + nomenclator	DEC	complete	–
augurelio1535	1533	German	substitution, no nomenclator	APT	complete	4363
cifuentes1534	1534	Spanish	syllabic nomenclator	DEC	complete	–
marechaux1520	1534	French	substitution, no nomenclator	KK	complete	1341
sperantio1534	1534	German, Latin	substitution, no nomenclator	APT	partial	4765
selve1535	1535	French	homophonic or simple + nomenclator	KK	partial	2524
charles1536	1536	Spanish	syllabic nomenclator	DEC	complete	–
rome1536	1536	French	homophonic or simple + nomenclator	KK	partial	695
caracciolo1537	1537	Italian	syllabic nomenclator	KK	complete	–
vatican5	1542	Italian	syllabic nomenclator	NS	none	6534
sadler1543	1543	English	undetermined	DEC	complete	–
malvezzi1548	1548	Latin	homophonic or simple + nomenclator	KK	partial	–
guiche1551	1551	French	substitution, no nomenclator	CT	partial	406
vargas1552	1552	Spanish	homophonic or simple + nomenclator	DEC	complete	–
santacroce1552	1553	Italian	homophonic or simple + nomenclator	DEC	complete	3933
harley1582r8499	1554	English	homophonic or simple + nomenclator	EK	partial	3152
renard1554	1554	French	undetermined	DEC	complete	–
r1212	1556	Latin	homophonic or simple + nomenclator	KK	complete	288
r1871	1558	Italian	homophonic or simple + nomenclator	DEC	complete	786
burghley1559	1559	English	code or codebook	DEC	complete	12
percy1559	1559	–	undetermined	NS	none	63
wod1568	1559	–	undetermined	KK	partial	39
memoire1560	1560	French	code or codebook	CT	complete	1715
throck1560	1560	English	undetermined	DEC	complete	–

target	year	language	system	method	extent	tokens
throckmorton	1560	English	homophonic or simple + nomenclator	DEC	complete	–
smith1562	1562	English	homophonic or simple + nomenclator	KK	complete	–
coligny1563	1563	French	homophonic or simple + nomenclator	DEC	complete	–
foix1563	1563	French	undetermined	DEC	partial	1007
lomer1563	1563	English	code or codebook	DEC	complete	7
middelmore1563	1563	English	code or codebook	DEC	complete	18
rennes1563	1563	French	homophonic or simple + nomenclator	KK	partial	2035
charlesixducroc	1565	French	homophonic or simple + nomenclator	DEC	complete	–
toledo1565	1565	Spanish	substitution, no nomenclator	APT	complete	1937
ducroc	1567	French	undetermined	NS	none	166
norreys1567	1567	English	homophonic or simple + nomenclator	KK	partial	7
nuncio1566	1567	Italian	homophonic or simple + nomenclator	KK	complete	7817
alessandrino1568	1568	Italian	homophonic or simple + nomenclator	KK	complete	10611
moray	1568	–	undetermined	NS	none	134
hamilton1569	1569	Scots	homophonic or simple + nomenclator	DEC	complete	–
throck1569	1569	English	homophonic or simple + nomenclator	KK	complete	–
norfolk1570	1570	Scots	homophonic or simple + nomenclator	KK	complete	–
randolph1570	1570	English	homophonic or simple + nomenclator	DEC	complete	–
birago	1571	Italian	undetermined	NS	none	483
burghley1572	1571	English	homophonic or simple + nomenclator	NS	none	4
r8356	1571	English	homophonic or simple + nomenclator	NS	none	11
mondoucet	1572	French	substitution, no nomenclator	APT	partial	7293
r8358	1572	English	homophonic or simple + nomenclator	NS	none	15
r8361	1572	English	homophonic or simple + nomenclator	NS	none	21
walsingham1572nov	1572	English	undetermined	NA	n/a	–
lanssac	1573	French	homophonic or simple + nomenclator	APT	complete	505
ormanetto1573	1573	–	undetermined	NS	none	1324
r8362	1573	English	homophonic or simple + nomenclator	NA	n/a	2
r8364	1573	English	homophonic or simple + nomenclator	EPT	partial	26
walsingham1572	1573	English	substitution, no nomenclator	NS	none	12
nevers1574	1574	French, Italian	substitution, no nomenclator	DEC	partial	–
chodkiewicz1575	1575	German	substitution, no nomenclator	DEC	complete	–
ormanetto1576	1576	–	undetermined	NS	none	389
ungnad1576	1576	–	homophonic or simple + nomenclator	NA	n/a	–
labbe1582	1577	French	undetermined	NS	none	911
sinzendorff1578	1578	–	undetermined	NA	n/a	–
dandini1580	1580	Italian	homophonic or simple + nomenclator	DEC	complete	1601
sauli1579	1580	Italian	homophonic or simple + nomenclator	DEC	complete	2392
r4157	1581	French	homophonic or simple + nomenclator	KK	complete	289
lodi1584	1584	Spanish	homophonic or simple + nomenclator	DEC	complete	307
harley1582r8505	1585	French	substitution, no nomenclator	CT	complete	3564
matignon1586	1585	French	homophonic or simple + nomenclator	KK	partial	12994
segur	1585	French	syllabic nomenclator	CT	complete	594
walsingham1585	1585	English	homophonic or simple + nomenclator	EPT	partial	–
wotton1585	1585	English	homophonic or simple + nomenclator	APT	complete	–
bergamo1585	1586	Italian	homophonic or simple + nomenclator	KK	complete	3216
clair357	1586	French	homophonic or simple + nomenclator	DEC	complete	–
harley1582r8500	1586	English	homophonic or simple + nomenclator	APT	complete	61
harley1582r8504	1586	English	homophonic or simple + nomenclator	APT	complete	7
stafford1586	1586	English	homophonic or simple + nomenclator	DEC	complete	28
guise1587	1587	French	homophonic or simple + nomenclator	KK	complete	2484
mercoeur1587	1587	French	homophonic or simple + nomenclator	KK	complete	952
needham1587	1587	English	substitution, no nomenclator	APT	complete	227
nevers1587	1587	French	homophonic or simple + nomenclator	EK	complete	27
vieuville1587	1587	French	homophonic or simple + nomenclator	NS	none	–
cobham1588	1588	English	homophonic or simple + nomenclator	APT	partial	–
harley287	1588	English	homophonic or simple + nomenclator	APT	partial	–
morosini1588	1588	Italian	homophonic or simple + nomenclator	KK	complete	234005
nevers1588	1588	French	homophonic or simple + nomenclator	APT	complete	344
r3708	1588	French	homophonic or simple + nomenclator	APT	complete	336
mendoza1589	1589	Spanish	syllabic nomenclator	DEC	complete	822
nevers1589†	1589	French	not a cipher	NA	n/a	–
orbais	1589	French	homophonic or simple + nomenclator	KK	partial	91
ottobon	1589	Italian	syllabic nomenclator	NS	none	1528
r2276	1589	French	homophonic or simple + nomenclator	KK	partial	3920
champagne1590	1590	French	homophonic or simple + nomenclator	APT	partial	196
gonzaga1590	1590	Italian	homophonic or simple + nomenclator	KK	complete	926
dinteville1592	1592	French	homophonic or simple + nomenclator	NS	none	–
lorraine1592†	1592	French	homophonic or simple + nomenclator	CT	partial	1109
maisse1592†	1592	French	homophonic or simple + nomenclator	DEC	complete	80
pelissier1592	1592	French	homophonic or simple + nomenclator	KK	partial	18451
stowe166	1592	English	homophonic or simple + nomenclator	CT	partial	–
lebel1593	1593	French	homophonic or simple + nomenclator	KK	complete	1935
nevers1593	1593	French	homophonic or simple + nomenclator	KK	partial	395
sega1593	1593	French	homophonic or simple + nomenclator	NS	none	–
sessa1593†	1593	Spanish	syllabic nomenclator	KK	partial	53
joyeuse	1594	French	syllabic nomenclator	NS	none	358
nevers1595	1595	French	homophonic or simple + nomenclator	NS	none	753
sloane3188	1595	English	not a cipher	NA	n/a	–
segura1596	1596	Spanish	substitution, no nomenclator	CT	complete	125
bethune	1601	French	syllabic nomenclator	APT	partial	2849
hesse1603†	1602	French	homophonic or simple + nomenclator	KK	complete	4100
breves1603	1603	French	homophonic or simple + nomenclator	DEC	complete	–
breves1610	1610	French	homophonic or simple + nomenclator	NS	none	100
louis1615	1615	French	homophonic or simple + nomenclator	DEC	complete	–

target	year	language	system	method	extent	tokens
andreae1616	1616	German	substitution, no nomenclator	APT	complete	156
cocquet	1616	French	undetermined	NS	none	–
r4817	1618	German	substitution, no nomenclator	EPT	complete	79
ferdinand1619	1619	German	homophonic or simple + nomenclator	CT	complete	2614
haga1620	1620	Dutch	homophonic or simple + nomenclator	KK	complete	–
sp106box10	1623	English, Italian	code or codebook	KK	partial	–
damiata1624	1625	Italian	homophonic or simple + nomenclator	DEC	complete	361
herbault1626†	1626	French	homophonic or simple + nomenclator	DEC	complete	–
warsaw	1627	Italian	syllabic nomenclator	CT	complete	465
boswell1628	1628	English	substitution, no nomenclator	KK	complete	442
r1862	1628	Italian	homophonic or simple + nomenclator	EPT	complete	1280
r1874	1628	Italian	homophonic or simple + nomenclator	DEC	complete	2245
riksarkivet1628	1628	German, Latin	homophonic or simple + nomenclator	APT	partial	3322
pallotto1629	1629	Italian	homophonic or simple + nomenclator	KK	complete	118931
richelieu	1629	French	homophonic or simple + nomenclator	APT	complete	421
ceva1632	1632	Italian	syllabic nomenclator	KK	partial	2400
ormonde	1634	English	homophonic or simple + nomenclator	APT	complete	59
harley286	1635	Latin	substitution, no nomenclator	CT	complete	281
marburg1635	1635	German	substitution, no nomenclator	CT	complete	590
rohan1636	1635	French	undetermined	NS	none	–
heusner1637	1637	German	homophonic or simple + nomenclator	KK	complete	259
kurtz1639	1639	German	substitution, no nomenclator	APT	partial	22913
baner1640	1640	German	homophonic or simple + nomenclator	APT	partial	288
harley7001	1641	English	homophonic or simple + nomenclator	EK	partial	254
boswell	1643	English	homophonic or simple + nomenclator	KK	complete	457
forster	1644	French	substitution, no nomenclator	KK	complete	207
lucca	1644	Italian	substitution, no nomenclator	EPT	complete	231
goring1645	1645	English	substitution, no nomenclator	KK	complete	338
hm1645	1645	English, French	homophonic or simple + nomenclator	KK	complete	–
rupert	1645	English	homophonic or simple + nomenclator	NA	n/a	93
rupert1645	1645	English	homophonic or simple + nomenclator	EK	complete	305
charlesi	1648	English	undetermined	NS	none	200
conti1649	1649	French	syllabic nomenclator	KK	complete	264
hamilton	1650	English	homophonic or simple + nomenclator	NA	n/a	102
bagno1652	1652	Italian	undetermined	NS	none	–
conley1652	1652	English	homophonic or simple + nomenclator	NS	none	23
urquhart	1652	English	code or codebook	KK	partial	349
bordeaux	1653	French	syllabic nomenclator	EK	complete	813
thurloe	1653	English	undetermined	NS	none	229
letellier	1657	French	undetermined	NS	none	202
hyde	1659	–	not a cipher	NA	n/a	41
r688	1664	Serbian	undetermined	NA	n/a	240
wesselenyi1664	1664	Latin	homophonic or simple + nomenclator	CT	complete	244
perwich	1670	English	transposition, polygraphic, polyalphabetic	DEC	complete	494
hernannunez1674	1674	Spanish	syllabic nomenclator	KK	complete	1716
colbert	1675	French	undetermined	NS	none	106
balbases1677	1677	Spanish	syllabic nomenclator	APT	complete	–
carpio1677	1677	Spanish	syllabic nomenclator	APT	complete	3504
ronquillo1676	1677	Spanish	syllabic nomenclator	KK	partial	23463
chaulnes	1690	French	syllabic nomenclator	NS	none	311
feuquieres	1690	French	code or codebook	EPT	complete	601
catinat1691	1691	French	code or codebook	KK	complete	12439
ilten1697	1697	French, German	code or codebook	DEC	complete	492
yard1699	1699	English	code or codebook	KK	complete	368
prussia1702	1702	French	undetermined	NA	n/a	56
stepney	1702	English	code or codebook	NS	none	24
bay1706	1706	Hungarian	syllabic nomenclator	EK	complete	414
papai1706	1706	Hungarian	syllabic nomenclator	KK	partial	17232
rakoczi1704	1706	French, Latin	syllabic nomenclator	EK	complete	5406
vetesi1705	1706	Hungarian	syllabic nomenclator	DEC	complete	–
henter1707	1707	Hungarian	homophonic or simple + nomenclator	DEC	complete	–
rakoczi1707	1707	French	syllabic nomenclator	EK	partial	1108
desmarets1710	1710	French	homophonic or simple + nomenclator	NS	none	471
geertruidenberg	1710	–	code or codebook	NA	n/a	–
anne1711	1712	English	code or codebook	DEC	complete	414
potocka1714	1714	French, Latin, Polish	homophonic or simple + nomenclator	CT	partial	1119
goertz1717	1717	German	homophonic or simple + nomenclator	NS	none	809
nunzio1718	1718	Italian	homophonic or simple + nomenclator	KK	complete	34243
bl32305	1719	French	code or codebook	NS	none	1677
windischgraetz1720	1720	German	syllabic nomenclator	KK	complete	415
colonia1721	1721	Italian	homophonic or simple + nomenclator	KK	complete	189
windischgraetz1721	1721	German	homophonic or simple + nomenclator	KK	partial	181
palm1727	1727	German	homophonic or simple + nomenclator	NS	none	185
visconti1727	1727	Italian	homophonic or simple + nomenclator	NA	n/a	193
r7707	1728	–	undetermined	NS	none	205
sp106r597	1731	French	homophonic or simple + nomenclator	NA	n/a	43
nunzio1736	1736	Italian	homophonic or simple + nomenclator	KK	complete	7386
ilten1743	1743	French	code or codebook	EK	complete	223
michell1751	1751	French	code or codebook	DEC	complete	912
hellen1752	1752	French	code or codebook	NS	none	2651
jones1753	1753	English	not a cipher	NA	n/a	–
kauderbach1754	1754	French	homophonic or simple + nomenclator	CT	complete	8512
affry1757	1757	French	homophonic or simple + nomenclator	APT	partial	9284
acciaiuoli1757	1758	Italian	homophonic or simple + nomenclator	KK	complete	8326
eichel1758	1758	–	homophonic or simple + nomenclator	NA	n/a	139

target	year	language	system	method	extent	tokens
swieten1757	1758	French	homophonic or simple + nomenclator	CT	complete	642
r1875	1761	Italian	substitution, no nomenclator	DEC	complete	1565
lucini1767	1767	Italian	homophonic or simple + nomenclator	KK	complete	152
destaing	1779	French	code or codebook	NS	none	217
deswart1782	1782	Dutch	code or codebook	EK	complete	1719
rechteren1785	1785	Dutch	homophonic or simple + nomenclator	DEC	complete	424
vanreede1787	1787	Dutch, French	code or codebook	KK	partial	1167
dedem1788	1788	Dutch	code or codebook	NS	none	941
vanreede1792	1792	French	code or codebook	KK	complete	83
hereditary1796	1795	Dutch, French	code or codebook	APT	complete	881
r1892	1795	Dutch, French	substitution, no nomenclator	CT	partial	1872
r2242	1795	Dutch	substitution, no nomenclator	EK	partial	1242
r2234	1796	French	transposition, polygraphic, polyalphabetic	KK	complete	1321
mellon136	1798	German	substitution, no nomenclator	KK	complete	1670
r1944	1801	Dutch	code or codebook	DEC	complete	110
r2232	1801	French	substitution, no nomenclator	KK	complete	1065
r1942	1803	Dutch	code or codebook	NS	none	286
toulon1803	1803	Dutch	code or codebook	NS	none	110
fagel1804	1804	French	homophonic or simple + nomenclator	NS	none	128
spaen1808	1808	–	code or codebook	NS	none	303
roell1809	1809	–	code or codebook	NS	none	2585
napoleon	1812	French	code or codebook	NS	none	326
castelcicala1816	1816	Italian	code or codebook	APT	partial	10397
loeschnen1816	1816	German	not a cipher	NA	n/a	–
zeschau1841	1841	French, German	syllabic nomenclator	NS	none	1896
soglia1848	1848	Italian	code or codebook	CT	complete	955
catokwacopa	1875	English	transposition, polygraphic, polyalphabetic	DEC	partial	441
adfgvx	1918	German	transposition, polygraphic, polyalphabetic	KK	complete	3966
blume	1937	Spanish	transposition, polygraphic, polyalphabetic	NS	none	615
dagapeyeff	1939	–	undetermined	NS	none	196
orpo1942	1942	German	transposition, polygraphic, polyalphabetic	NS	none	1191
censorship	1943	German	substitution, no nomenclator	NS	none	–
abwehr	1944	–	undetermined	NS	none	924
enigma	1945	German	transposition, polygraphic, polyalphabetic	NS	none	97
blancmesnil	–	French	undetermined	NA	n/a	–
egmond	–	French	substitution, no nomenclator	CT	complete	774
fr3029	–	French	undetermined	KK	complete	–
fr3029f134	–	French	homophonic or simple + nomenclator	KK	complete	926
fr3092	–	French	homophonic or simple + nomenclator	KK	complete	4521
mercoeur1586	–	French	homophonic or simple + nomenclator	NS	none	217
r1818	–	–	undetermined	NS	none	355
r8870	–	–	code or codebook	NS	none	901
r8871	–	–	undetermined	NS	none	77
sp106r927	–	French	substitution, no nomenclator	CT	partial	–
sp53	–	–	undetermined	NS	none	1235
superscript	–	Latin	undetermined	NA	n/a	–